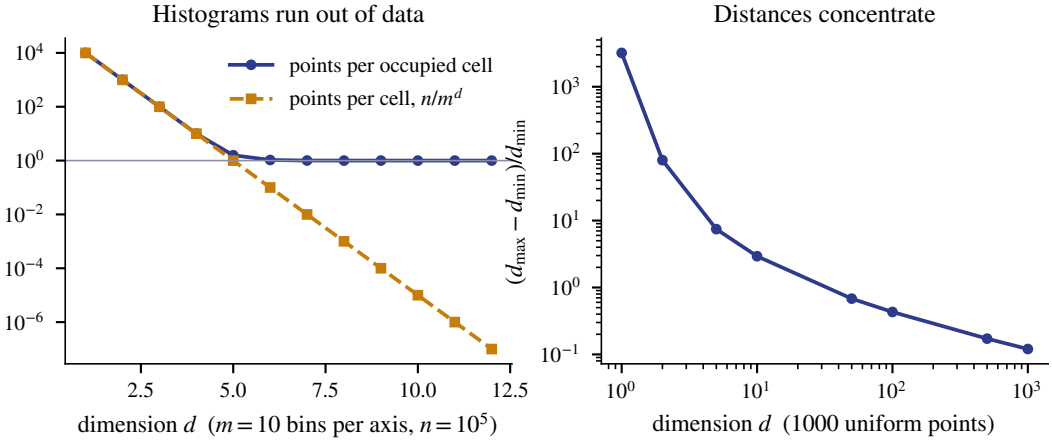
Histograms run out of data
points per occupied cell
points per cell, $n/m^d$
$10^4$
$10^2$
$10^0$
$10^{-2}$
$10^{-4}$
$10^{-6}$
2.5
5.0
7.5
10.0
12.5
dimension $d$ ($m = 10$ bins per axis, $n = 10^5$)
Distances concentrate
$(d_{\max} - d_{\min})/d_{\min}$
$10^3$
$10^2$
$10^1$
$10^0$
$10^{-1}$
$10^0$
$10^1$
$10^2$
$10^3$
dimension $d$ (1000 uniform points)